



Mid-Tier Defense Against Hypersonic Glide Vehicles During Cruise

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A detailed conceptual missile system design and analysis was conducted to develop a single-stage hit-to-kill missile system featuring a blast fragmentation lethality enhancer to counter threat hypersonic glide vehicles (HGV) during cruise. The missile system is capable of defending a 7 km radius protection area against non-maneuvering and maneuvering hostile HGVs at an altitude ranging from 120-140 kft with a cruise Mach number of 12-15. A comprehensive missile system design methodology that utilizes an iterative design procedure to fulfill provided and derived requirements is discussed in detail. This includes the interdisciplinary missile design analysis and procedures, and description of detailed analysis tools developed to expedite the missile design process. To complete the analysis, details about the missile subsystem components are provided with an elaborate inboard layout of the conceived missile system. The content presented in this paper is a section of the report submitted as part of the AIAA Missile Systems Technical Committee 2016-2017 Design Competition.

Nomenclature

a	=	empirical constant
a_m	=	normal acceleration command
A_b	=	surface area of solid rocket motor
A_e	=	nozzle exit area
A_p	=	port area
A_t	=	nozzle throat area
A_{z_m}	=	missile sensor acceleration
C/M	=	charge to mass ratio of warhead to fragment
c^*	=	characteristic velocity
C_a	=	axial force coefficient in stability axis
C_l	=	coefficient of lift
C_d	=	coefficient of drag
C_m	=	coefficient of moment in stability axis
C_{mq}	=	coefficient of moment due to pitch rate
$C_{m_{cmq}}$	=	derived coefficient of moment from the moment coefficient due to pitch rate
C_n	=	normal force coefficient
C_p	=	heat capacity at constant pressure
C_x	=	axial force coefficient in body axis
C_z	=	normal force coefficient in body axis
C_D	=	discharge coefficient
C_F	=	thrust coefficient
D_f	=	drag force

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f	=	width of groove in star shape
F_x	=	axial force at the center of gravity
F_z	=	normal force at the center of gravity
g	=	acceleration due to gravity
H_c	=	enthalpy of chamber gases
H_e	=	enthalpy of exhaust gases
HGV	=	hypersonic glide vehicle
I_{sp}	=	specific impulse
I_{yy}	=	missile moment of inertia about pitch axis
K	=	ratio of surface area to port area
L	=	length of the sub-phase boost
L_f	=	lift force
L_m	=	length of missile
L_t	=	length of target HGV
m	=	instantaneous missile mass
$\dot{m}$	=	propellant mass flow rate
M	=	moment about the center of gravity
n	=	burn rate exponent
N	=	number of grooves for star shape
N'	=	proportional navigation constant
p_c	=	chamber pressure
p_e	=	exit pressure
p_l	=	pressure at the inner region of boost phase
q	=	missile pitch rate
$\dot{q}$	=	rate of missile pitch rate
r	=	burn rate of propellant
R	=	range between target and missile
R_o	=	radius of missile
Re	=	Reynolds number per unit foot
S_1	=	surface area 1
S_2	=	surface area 2
S_{ref}	=	missile reference area
t	=	burn time
T	=	freestream temperature
t_c	=	time of closest approach
T_c	=	chamber temperature
T_e	=	exit temperature
$\dot{u}$	=	missile longitudinal acceleration
U	=	missile longitudinal velocity
v_e	=	exit velocity
V	=	resultant missile velocity
V_c	=	chamber volume
V_t	=	target velocity
y	=	depth of burnt propellant
w	=	missile normal velocity
α	=	angle of attack
δ	=	tail fin deflection
ΔH	=	heat of formation
$\bar{f}$	=	area ratio between throat and port
$\dot{\lambda}$	=	rate of change of line-of-sight
γ	=	heat capacity ratio
γ_m	=	missile flight path angle
γ_t	=	target flight path angle
ω_n	=	angle between axis and normal to fragment
μ	=	absolute viscosity of air

θ = missile pitch attitude
 ρ = density of propellant
 ρ_a = density of air

I. Introduction

A. Background

THE concept of hypersonic powered and boost-glide weapon systems has received interest in recent decades. Notable experimental research efforts within the U.S. include the X-43, the X-51, the Falcon Hypersonic Technology Vehicle 2 (HTV-2), and the Advanced Hypersonic Weapon (AHW) [1–4]. The X-43 served as an unmanned experimental hypersonic aircraft that was launched in 2004 to test various aspects of airbreathing hypersonic flight. Similarly, the X-51 was an unmanned research scramjet aircraft, which underwent four flight tests from 2010–2013. The primary purpose of the X-51 project was to demonstrate the capability of sustained hypersonic powered flight by means of an airbreathing scramjet propulsion system. Following successful testing, the X-51’s technology was proposed for use in the High Speed Strike Weapon (HSSW), intended to provide the basis of future hypersonic expendable missiles and reusable systems [5]. Further, the HTV-2 served as an advanced experimental hypersonic glide vehicle (HGV) to provide enhanced testing and evaluation of HGV aerodynamics, aerothermal effects, and critical guidance, navigation, and control [3]. Two flight tests were conducted throughout 2010–2011, upon which the HTV-2 was rocket launched atop a Minotaur IV launch platform, and then deployed as a re-entry glide vehicle. The two flight tests demonstrated hypersonic glide velocities of approximately Mach 5–22 with an equilibrium glide aerodynamic efficiency of roughly 2.6, and an operational glide range of approximately 3,800 miles [6]. Ideally, future HGVs could demonstrate performance characteristics for a Prompt Global Strike (PGS) capability. In principle, a PGS weapon system is classified as a precision-guided system that is capable of delivering a weapon strike anywhere in the world within one hour [7]. In 2011, as part of the U.S. PGS program, the AHW was flight tested and demonstrated the capability of striking a target 2,300 miles downrange in under 30 minutes [8]. Although no deployable U.S. hypersonic weapon system is known to exist to date, extensive research and development efforts are now focused on integrated system and subsystem development.

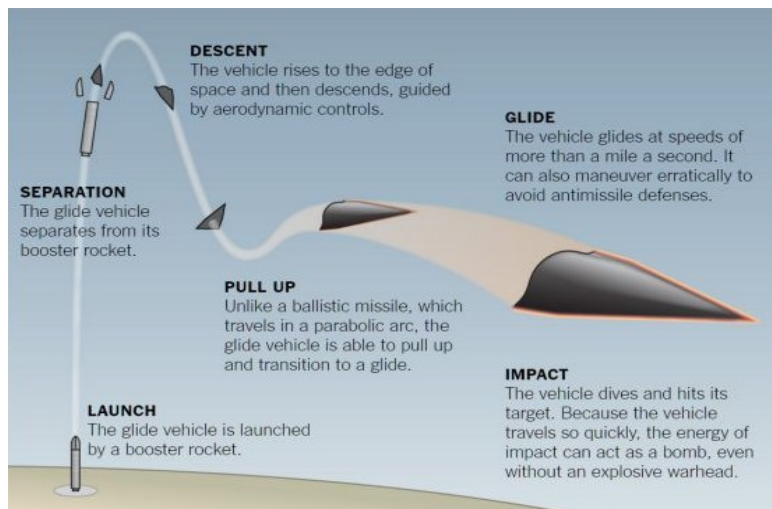
Research efforts within the U.S. have made the concept of high-precision hypersonic weapon systems realizable and possibly capable of deployment within a few decades. the roadmap for hypersonic weapon systems within the U.S. is aimed at high-speed strike weapons capable of rapid strike from standoff by the mid 2020’s, tactical strike weapon systems featuring advanced intelligence, surveillance, and reconnaissance (ISR) equipment by the mid 2030’s, and eventually reusable and persistent weapon systems capable of penetrating adversary territory and conducting ISR or weapon strikes by the mid 2040’s [9]. In addition to U.S. research efforts, the goal of deploying hypersonic weapon systems has garnered interest globally. Recently, Russia and China have engaged in independent research efforts concerning the development of hypersonic weapon systems in the form of HGVs. As indicated in Table 1, Russia has recently been developing HGV weapon technologies with experimental testing of their YU-71/YU-74, with the most recent testing conducted in 2016. Similarly, China has independently been developing HGV weapon technologies with their DF-ZF/DF-17 and Xingkong-2 experimental HGVs, with the most recent flight testing over the course of 2014–2017, and August 2018 respectively. As a result, increasing foreign interest in HGV development demands a missile defense system tailored for countering the potential HGV threat.

Table 1. Comparison of recent and current global HGV weapon system development.

		
Image: wikipedia.org	Image: sputniknews.com	Image: wikipedia.org
U.S. – Falcon HTV-2 ^[1]	Russia – YU-71^[3], YU-74	China –DF-ZF/Xingkong-2 ^[11]
		
- Role: Exp. R&D. - 1st flight test: Apr. 2010 - Last test: Aug. 2011 - Tests to date: 2 - Max Mach number: 20	- Role: Exp. R&D. - 1st flight test: 2015, 2016 - Last test: April 2016 - Tests to date: ~ 2 - Max Mach number: ~10	- Role: Exp. R&D. - 1st flight test: Jan. 2014 - Last test: August 2018 - Tests to date: 9 in total - Max Mach number: 6-10

B. Motivation

In principle, the basic operation of an HGV consists of a launch platform vehicle, which houses the glide vehicle and boosts the vehicle to an exo-atmospheric altitude. At an intermediate point, the glide vehicle separates from the booster and continues on a ballistic trajectory through an apogee point and further into a re-entry descent. After re-entering the atmosphere, the HGV reaches a point during descent whereby the vehicle pulls up using aerodynamic controls. At this point, the HGV departs from the predictable parabolic path of a conventional ballistic missile and enters an equilibrium glide phase. During the glide phase, the HGV decelerates from roughly Mach 20 to Mach 5 over the course of traversing nearly 12,000 miles. At the end of the glide phase, the HGV can either act as a kinetic strike weapon with or without a warhead, or potentially even deliver a warhead to a designated target location. Figure 1 shows schematically the basic operation of an HGV flight envelope from the launch location to just before impact or warhead delivery [12].

**Figure 1. Flight envelope of typical HGV operation [12].**

The HGV provides superior threat capability compared to a conventional intercontinental ballistic missile (ICBM). The enhanced threat capability stems from the fact that the HGV does not follow a predictable parabolic decent trajectory like that of an ICBM, is maneuverable during the mid-course and terminal phase, and has the potential of gliding for long duration under the detection region of modern, ground-based radar systems. Additionally, due to the lifting body configuration design of HGVs, the HGV can potentially traverse longer distances in less time than the typical ICBM. These factors make defending against HGV threats especially challenging, such that operational HGVs could have the capability of penetrating current ballistic defense systems such as the Terminal High Altitude Area Defense (THAAD) and Ground-based Midcourse Defense (GMD) [6].

Since the HGV glides within the atmosphere at very fast glide speeds, it is likely that they can counter ballistic missile defense systems using exo-atmospheric kinetic kill vehicles [13]. Similarly, although exo-atmospheric vehicles have successfully demonstrated endo-atmospheric intercepts using a kinetic kill vehicle, it is unknown how they would fair against a maneuvering HGV at lower altitudes, given the added complexity of atmospheric frictional heating and the reduced efficiency of the exo-atmospheric vehicle's infra-red seeker [6, 14]. Lastly, it is evident that conventional surface-to-air missile (SAM), do not deliver the required altitude to intercept an incoming HGV at its typical cruise glide altitude. Thus, it is clear that a need exists to counter the HGV threat.

II. Missile System Design Requirements

In response to the emerging HGV threat and the need for a capable counter HGV missile defense system, detailed design requirements and objectives of the proposed missile defense system were formulated. This section details and summarizes the relevant, major design requirements and presents further derived requirements based on the anticipated required performance of the proposed counter HGV missile defense system.

A. Formulated Requirements

The overall design requirement is to develop a ground-based, counter-HGV missile defense system that can defend and protect a 7 km (4.35 mile) radius against hostile HGVs during their cruise phase. The major design requirements are summarized into four primary categories. The two categorized technical requirements that are addressed in this paper are summarized below while all the major requirements are presented in Table 3.

1. Storage

The missile system must be capable of storage for 20 years without maintenance and engage a total of 100 HGVs over the course of its lifetime. The system lifetime consists of 100 months deployed and 140 months in storage.

2. Threat

The hostile HGVs may range from 2,000–5,000 lbs with a nominal operating altitude between 120–140 kft and approach with Mach numbers ranging from 12–15. The system must feature the necessary systems and subsystems to eliminate incoming maneuvering or non-maneuvering HGV threats.

Table 2 below provided a condensed summary of the major requirements, and Figure 2 schematically showcases these requirements.

Table 2. Summary of counter HGV missile defense system requirements.

	Requirement	Metric
Cost	Cost of System	Minimize Cost
Storage	System Life Span	Threat HGVs Total: 100 Avg. Annual Threat HGVs: 5 Life Cycle: 20 years (100 months deployed, 140 months stored)
Threat	Detect HGV Threat	Range: > 7 km
	Destroy HGV Threat	HGV Weight: 2000-5000 lbs HGV Cruise Speed: Mach 12-15 HGV Cruise Alt: 120-140 kft
	Protect Area	Surface-to-Air Launch 7 km radius
	Destroy Multiple HGVs	Non-Maneuvering/Maneuvering HGVs Wave: 3 HGVs at 15 s intervals Multiple Waves: 1 hr intervals
CONOPS	Quick Setup and Operation	Setup and Operation: < 90 min
	Operation Conditions	Weather: Any HGV operating conditions Environment: Urban and Rural
	Ability to be Transported	Any terrain capable of heavy duty tactical trucks can normally access

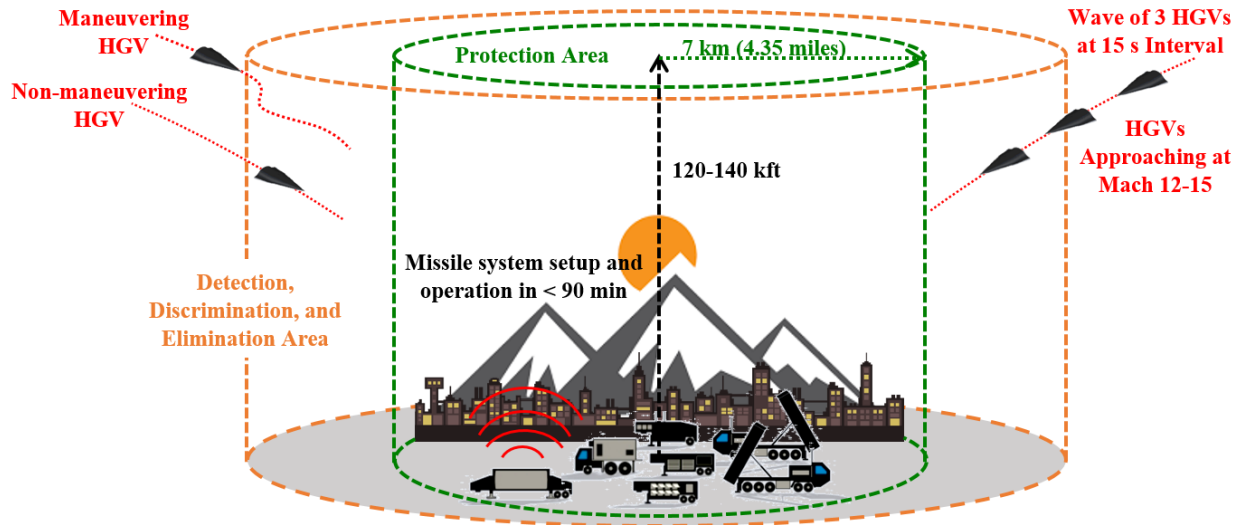


Figure 2. Schematic of missile system defense requirements.

B. Derived Requirements

Based on the technical requirements, additional mission requirements can be derived that aid in identifying a feasible design solution subject to anticipated missile system performance. Given the altitude at which the target HGVs could potentially approach at, a detection distance can be estimated based on geometric constraints of ground based radar systems.

Figure 3 schematically describes the detectable and shadow zones of a ground based radar system due to the curvature of the earth. Based on geometric constraints, the detectable distance considering the target HGV altitude is given by,

$$R_{\text{Radar}} = 2R_e \text{Alt}_{\text{HGV}} + \sqrt{2h_{\text{Radar}}R_e} \quad (1)$$

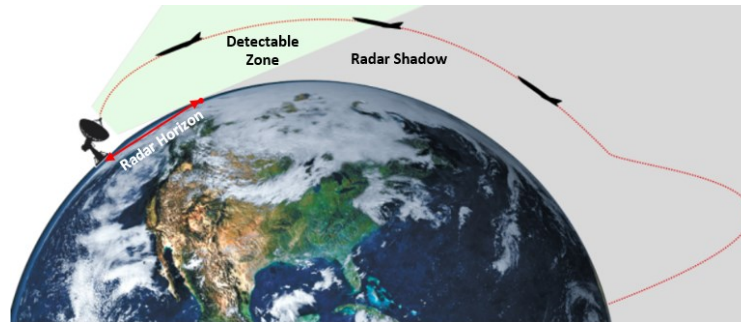


Figure 3. Schematic of detectable and shadow zones of a ground based radar system.

Thus, given a target HGV altitude of 120–140 kft, it is estimated that the maximum theoretical detection distance of a ground based radar system is approximately 420–460 miles. This yields a total closing time of roughly 165 seconds for an incoming target HGV after the time of maximum detection, assuming the radar system is at the location of the protection area. Moreover, based on practical missile defense system performance metrics, if it is assumed that a counter HGV missile is launched at the location of the ground based radar system and accelerates to Mach 6 at the terminal phase, a propellant burn time of approximately 110 seconds is required to intercept the target HGV at a downrange distance of approximately 150 miles. It is noted that such a burn time requirement is not feasible for a single stage missile system so that a multi-stage missile system is most likely required. Further, if a burn time of approximately 60 seconds is considered, a realistic maximum burn time of a single-stage missile, then a missile system could theoretically intercept an incoming HGV at roughly 40–50 miles downrange with an approximate intercept Mach number of 6. Since these performance parameters satisfy the requirements, it is feasible to pursue the design of a single-stage missile system to minimize the overall cost of the system.

From an energy balance regarding the available specific energy from the propulsion system and a desired intercept kinetic energy and gravitational potential energy, an estimate can be established for the propellant mass fraction of the missile system. Assuming that a solid rocket propulsion system can deliver a specific impulse of 260 seconds, the propellant mass fraction required to intercept an HGV at 140 kft at Mach 6 is approximately 0.60. This estimate assumes a 25 percent loss in available specific energy due to aerodynamic drag, gravitational loss, and overall propulsion system efficiency. In summary, for a single-stage missile system, Table 3 details the derived performance requirements based on the threat requirements and estimated feasible missile system constraints. It can be further concluded from these derived requirements that a single-stage missile system would need to satisfy an engagement protocol sequence within roughly 140 seconds after the detection from a ground based radar system.

Table 3. Realizable derived performance metrics for a single stage missile defense system.

Derived Requirement	Metric
Maximum Detection Distance	$R_{\text{Radar}} \sim 440$ miles
Intercept Velocity	$\text{Mach} \sim 6 / V \sim 6,200$ ft/s
Missile Range	$R \sim 40\text{-}50$ miles
Missile Altitude	$\text{Alt} \sim 140$ kft / ~ 27 miles
Time-to-Target	$t \sim 60$ s
Propellant Mass Fraction	~ 0.6

III. Missile System Design Methodology

In order to appropriately design a missile defense system that is capable of countering hostile HGVs during their cruise phase, the design methodology of Fleeman is utilized [21]. A flow chart of this specific design methodology is outlined in Figure 4.

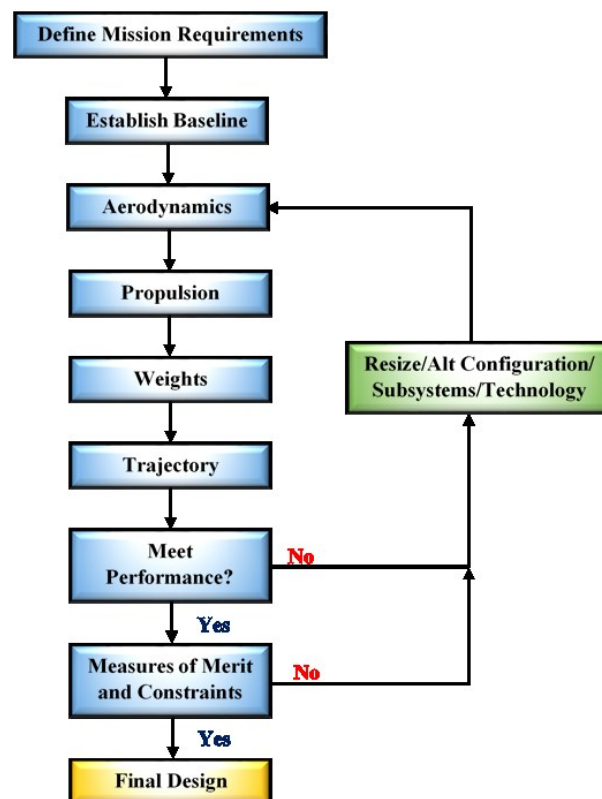


Figure 4. Conceptual iterative design methodology flowchart [21].

After the mission requirements for a missile system are defined, the iterative design methodology depicted in Fig. 4 presents a synthesis and systems engineering approach to rapidly iterate a baseline missile system to satisfy and meet the required performance metrics and measures of merit for a specified mission statement. For the current design, an analysis of alternative baseline missile configurations and mission statements are not considered due to

time constraints for the overall study. In order to initiate the design methodology, it is required to assess the mission and derived performance requirements to determine which existing missile system best provide the required performance. The advantages of using a baseline missile include the prior consideration of balanced systems engineering for the subsystems and use of an accurate benchmark based on existing data [21]. From this point, changes are made in the configuration sizing, missile aerodynamics, propulsion, weight, subsystems, and flight trajectory to ultimately satisfy the mission requirements. Changes in the missile geometry configuration drive the aerodynamics of the missile, which then necessitates iterative analysis of the other design disciplines. Hence, once a baseline is established, the baseline aerodynamics of the missile is analyzed. The aerodynamic data is then used to size the missile propulsion system to deliver performance aimed at satisfying the required range and time-to-target. With the propulsion system sized, an estimate of the missile weight and mass properties is then conducted. With knowledge of the missile aerodynamics, propulsion system, and weight properties, a simulated trajectory is computed to investigate how well the missile performs against an incoming HGV threat. After computing several trajectories, the nominal terminal velocity, range, altitude, and time-to-target are compared against the provided and derived mission requirements. If the missile system fails to satisfy any of the required performance metrics, the missile system is altered by means of resizing, alternate configurations, and selection of different subsystems and technologies, until the missile satisfies the required performance. This design process may require several iterations depending on the complexity of the required performance and capability of the selected baseline missile. Lastly, once the missile system satisfies the required performance metrics, analysis of the measures of merit and systems engineering and constraint requirements is conducted. The following subsections provide detailed discussion regarding the manner in which the analyses for each design discipline were conducted.

A. Baseline Missile

In order to initiate the iterative missile design procedure previously outlined, it is required to investigate the performance capability of several existing missile defense systems. The ultimate goal of selecting a baseline missile is to select a single missile system that provides performance metrics that are comparable to those required to counter an HGV. The GMD is designed for exo-atmospheric kills against ballistic missiles, and the terminal velocity far exceeds that required to effectively intercept an HGV. Therefore, the GMD is not considered as a baseline configuration. The THAAD provides a range, altitude, and intercept velocity that exceeds that required to intercept the required target HGV, however, the THAAD system has demonstrated the capability of intercepting endo-atmospheric targets despite being equipped with an exo-atmospheric kill vehicle [14]. In the case of the PAC-2, the range is sufficient, however the altitude requirement would necessitate a larger booster stage. With PAC-3, the missile range and altitude are nearly half of that anticipated to counter an HGV, and thus will not be pursued as a baseline. Considering the SM-2, though the range provided is very close to that required based on the derived performance, the required altitude would warrant a larger booster. Lastly, considering the multistage SM-3, the provided range, altitude, and intercept velocity far exceed that required for the current missile performance. Therefore, the SM-3 will not be considered as a baseline missile.

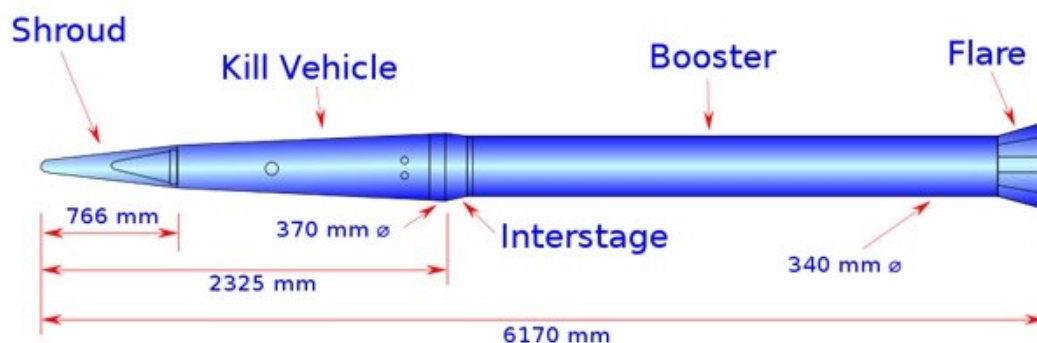


Figure 5. Schematic of THAAD missile configuration and kinetic kill vehicle [22].

From the above discussion and reasoning, the THAAD system is considered an appropriate baseline model. Figure 5 shows a dimensioned schematic of the THAAD missile configuration. The THAAD missile in its current form would not satisfy the derived missile performance requirements, and thus several modification will be pursued to establish a baseline missile configuration. Due to the high altitudes at which the THAAD is designed to intercept incoming targets, THAAD makes use of attitude control motors and thrust vector control, as the thin atmosphere at these altitudes renders aerodynamic control surfaces ineffective. However, in the case of the current mission, the

maximum altitude requirement is approximately 27 miles compared to the 90 miles for THAAD. At an altitude of 27 miles, a missile moving at Mach 6 in the terminal phase would have roughly 0.75 psi of dynamic pressure to be used with aerodynamic control surfaces. Using linear wing theory plus Newtonian impact theory [21],

$$C_{N,surface} = \left[\frac{4 \sin \delta \cos \delta}{\sqrt{M^2 - 1}} + 2 \sin^2 \delta \right] \frac{S_{surface}}{S_{ref}} \quad (2)$$

A control surface with an area of 1.5 that of the missile frontal area and a deflection of 0.17 radians would yield a normal force coefficient of 0.264. Further, with two control surfaces acting in a given flight direction this would yield a control surface force at Mach 6 and 27 miles of altitude of roughly 60 lbf. Although this control surface force is not considerably large, a tail control missile configuration is worth investigating due to the simplicity of the design and cost requirements compared to an actuated nozzle design with attitude control motors. Therefore, for the baseline missile a conventional cruciform tail design will be considered with a 5% double wedge supersonic airfoil with a taper ratio of 0.25, 50 degree leading edge sweep, and a control surface area of $1.59S_{ref}$. The addition of a cruciform tail on the baseline shifts the aerodynamic center of the missile further aft, and the flare stabilizer of the THAAD is removed and replaced by a 5 deg boat tail to reduce base drag during flight. For the baseline configuration the nose shape is approximated as a tangent-ogive with a fineness ratio of 6.8. Nose bluntness is not considered on the baseline and will be investigated in further iterations after analyzing the aerodynamic heating at the nose stagnation point. Lastly, on the basis of cost, it is desired to investigate the possibility of replacing a kinetic kill vehicle with a warhead lethality enhancer and operate the missile as a single stage hit-to-kill system. Figure 6 below shows a schematic of the baseline missile configuration, and Table 4 includes the baseline configuration geometry specifications.



Figure 6. Baseline missile configuration.

Table 4. Baseline missile geometry.

Component	Dimension
Length Nose	7.63 ft
Diameter of Nose	1.12 ft
Length of Centerbody	12.61 ft
Diameter of Centerbody	1.12 ft
Boat Tail Length	1.00 ft
Boat Tail Angle	5°
Airfoil	Double-wedge
Airfoil Thickness	5 %
Root Chord	2.00 ft
Taper Ratio	0.25
Semi-Span	1.81 ft
Leading Edge Sweep	50°
Fin Area	1.56 ft ²
Reference Area	0.99 ft ²

B. Aerodynamics

The aerodynamics methodology consists of utilizing Missile DATCOM, an aerodynamic design tool with suitable predictive accuracy for conceptual and preliminary design of a wide variety of conventional missile configurations. For this study, the 1997 Missile DATCOM Fortran code is used. [23, 24]. Such a code permits rapid aerodynamic sweeps of numerous missile configurations which aids in expediting the iterative analysis of the design methodology. For the purposes of this study, it was desired to have the Missile DATCOM Fortran source code

executed from within a MATLAB environment to streamline the aerodynamics analysis and data storage. Additionally, this allowed for a platform to establish a smooth transition of aerodynamic data to other design disciplines, including a trajectory analysis.

In order to execute Missile DATCOM from MATLAB, an interface was developed using MATLAB scripts to provide the necessary input data structures for Missile DATCOM. This included the specification of the missile geometry, including details of the nose, center-body, boat-tail, and fin layout. Additionally, a flight condition is specified subject to a flight Mach number, angle of attack, control surface deflection, and Reynolds number per unit length. In this case, depending on the flight Mach number, MATLAB switches the aerodynamic methods to ensure the proper aerodynamic methods are being utilized for a given flight regime. As an example, regarding the body normal force, the second-order shock expansion subroutine is utilized over the Mach number range $1.2 < M < 5$, upon which MATLAB switches the executed subroutine to modified Newtonian for Mach numbers greater than 5. The missile geometry and flight condition are stored in data structure files that are then passed to a secondary MATLAB function that modifies the required input files to execute the Missile DATCOM source code. This secondary function modifies a standard for005 input file by scanning the input file for placeholder strings and then replacing these strings with the proper geometry and flight condition input data. Once the input file is properly modified, Missile DATCOM is executed using system commands within MATLAB. In order to obtain the desired aerodynamic data for this study, the longitudinal static and dynamic aerodynamics are computed including C_N , C_A , C_M , X_{CP} , the associated pitch rate derivatives, and the missile trim condition. After Missile DATCOM generates the missile configuration aerodynamics data in the for006 file, and the trim calculations from a control surface deflection sweep in the for003 file, an additional MATLAB function opens these files and extracts the desired aerodynamic data. The data is then stored in a 4-D MATLAB structure variables that serves as aerodynamic lookup tables for the 3-DOF trajectory analysis and for reference regarding other design disciplines. In order to sweep an adequate range for this study, each missile configuration was swept over a flight Mach number range of $0.2 < M < 7.0$, an angle of attack range of $-12^\circ < \alpha < +12^\circ$, a control surface deflection range of $-20^\circ < \delta < +20^\circ$, and a Reynolds number per foot of $10^5 < Re_L < 10^9$. Figure 7 is a flowchart that details the aerodynamics analysis interface used to execute Missile DATCOM from within MATLAB. Additionally, Figs. 8–11 are a summary of the aerodynamic data output from Missile DATCOM.

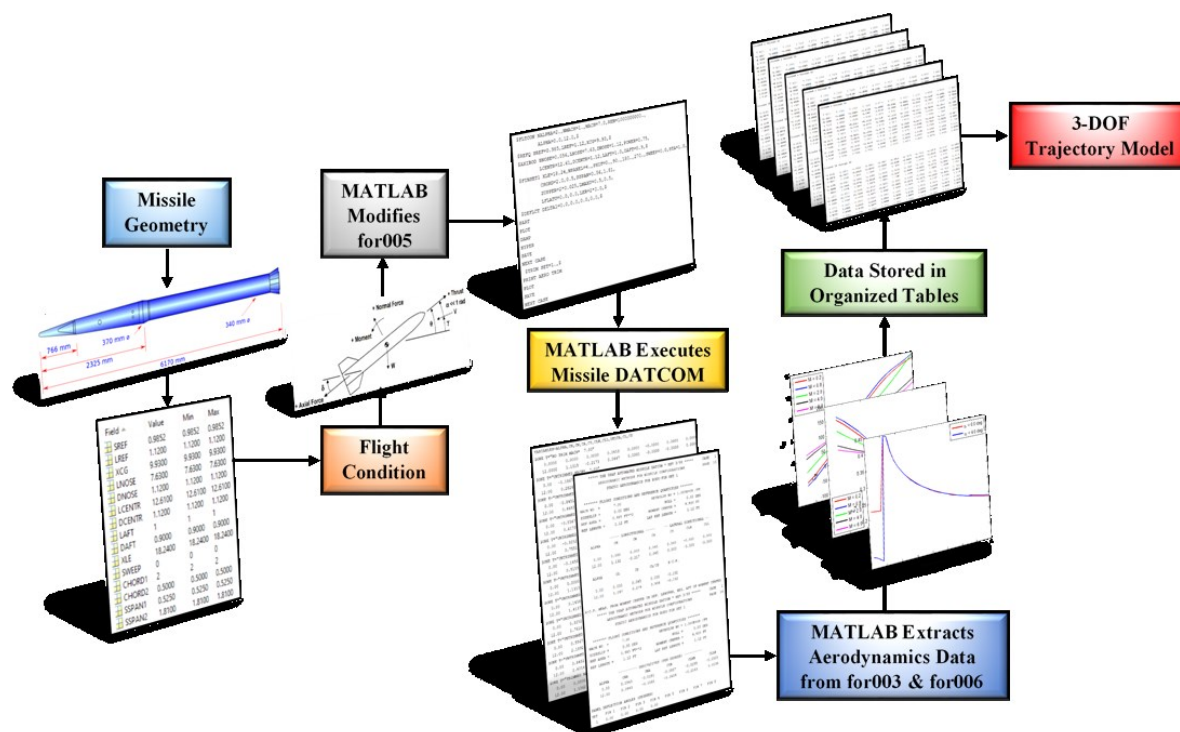


Figure 7. Flowchart of MATLAB/Missile DATCOM interface for missile aerodynamics analysis.

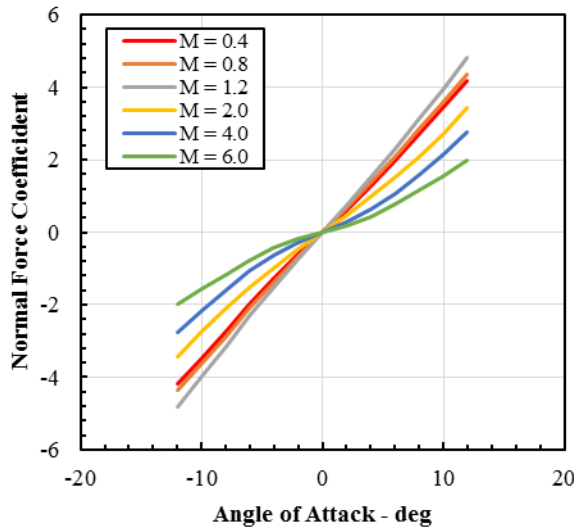


Figure 8. Baseline C_N vs angle of attack.

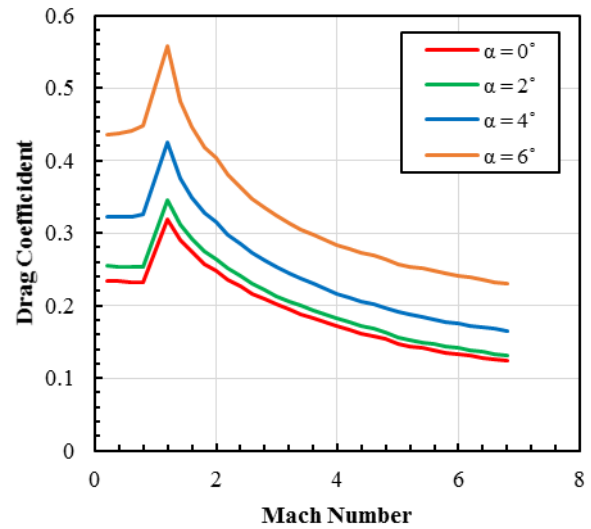


Figure 9. Baseline C_D vs Mach Number.

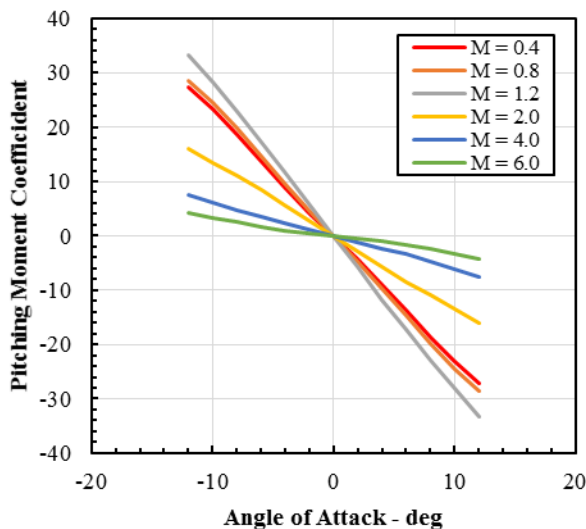


Figure 10. Baseline C_M vs angle of attack.

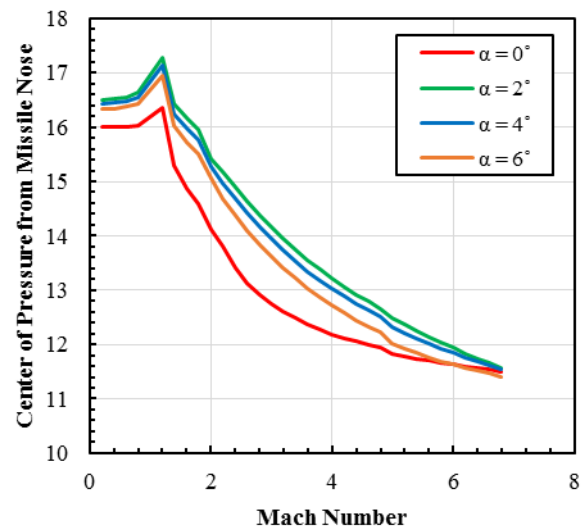


Figure 11. Baseline X_{CP} vs Mach number.

C. Propulsion

The capability to meet the derived mission range and altitude requirements is driven by missile propulsion performance. Due to storage and cost constraints for the missile system, it is desired to investigate the use of a solid propellant propulsion system, compared to liquid or hybrid propellant systems. Worth considering additionally is that solid propellant systems are less complex to operate than their liquid and hybrid propellant counterparts.

The propulsion requirement for this design is governed by the length and diameter of the missile, burn-time, missile velocity at the terminal phase, altitude change with burn time, storage conditions, and cost. These requirements are used to select and characterize the propellant and design the grain. Desired solid propellant characteristics to satisfy the mission requirements consists of high deliverable specific impulse, adequate mechanical and thermal properties to withstand a range of environmental conditions, low-hazard manufacturing and easy handling characteristics, characterize long term storage capability and suitable aging characteristics, low material and manufacturing cost. Furthermore, it is desired that the propellant possesses desirable burning characteristics

such as, low pressure exponent for propellant burning rate, low temperature sensitivity, minimum erosive burning, suitable ignition behavior, high reproducibility in ignition and burning characteristics, and high propellant density [25, 26].

These propellant characteristics help in selecting and tailoring the solid propellant design. For the desired propulsion requirement, high specific impulse propellants are needed that could withstand the storage limit of 20 years in extreme conditions and operable at any environmental conditions. These propellant composites generally consist of an energetic binder and a metallic fuel to enhance the performance of the propellant. Among these compositions, ammonium perchlorate composite (APCP consisting of ammonium perchlorate as an energetic material, hydroxyl-terminated polybutadiene as a binder and aluminum as a metallic fuel) and ammonium dinitramide composite (ADNCP consisting of ammonium dinitramide, hydroxyl-terminated polybutadiene and aluminum) are considered for calculating the propellant performance. These propellant composites have high specific impulse and low material and manufacturing cost with comparatively high density, desired burn rate and sufficient physical properties to withstand the storage and environmental conditions. Also, APCP has been used in many different missiles and rockets, which aids to emphasize its reliability. The selected propellants are evaluated by finding the theoretical propellant performance, which is needed to characterize, design the grains, and obtain the desired thrust profile for the mission.

The theoretical propellant performance calculations are based on two methods of deriving chamber composition: the thermodynamic equilibrium constant approach, and the minimization of Gibbs free energy approach. In the present work, the first method is used. The validity of the calculation depends primarily on the degree to which the equilibrium chamber and exhaust composition can be defined [28].

Assumptions made for the thermodynamic equilibrium method are: the gaseous species obey perfect gas law, Dalton's law is applicable, molar enthalpies and entropies are independent of pressure, steady-state condition exist during combustion and expansion, condensed phase occupy a negligible volume, combustion occurs at constant pressure, homogeneous mixing occurs, isentropic, frictionless, one-dimensional flow occurs in the nozzle, and the particles are thermodynamically macroscopic [29].

Using the macroscopic, steady-state flow process described in [29], the specific impulse equation is given by,

$$Isp = 6.940 \sqrt{H_c - H_e} \quad (3)$$

Similarly, the thrust coefficient and nozzle area ratio are given by,

$$C_f = \sqrt{\frac{4\gamma^2}{\gamma^2 - 1} \left(\frac{2}{\gamma + 1} \right)^{\frac{2}{\gamma-1}} \left(1 - \frac{T_e}{T_c} \right)} \quad (4)$$

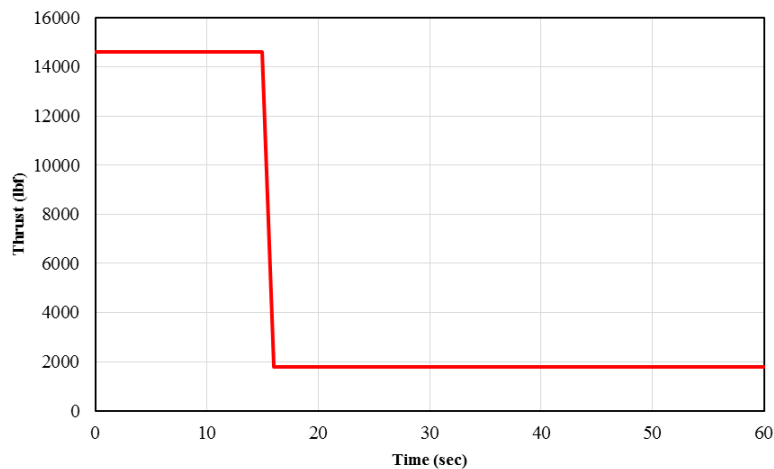
$$\frac{A_e}{A_t} = \frac{\gamma}{C_f} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{\gamma-1}} \left(\frac{p_c}{p_e} \right)^{\frac{1}{\gamma}} \quad (5)$$

The solution procedure is by trial-and-error, which requires a chamber enthalpy and nozzle entropy balance at an assumed chamber and exit temperature. First, a combination of propellant composition, combustion pressure, and exhaust pressure are selected, then the products of combustion are qualitatively selected, and independent mass balance and chemical equilibria equations are written to allow simultaneous solution of the system. Next, the combustion temperature is assumed and the composition of the chamber products are calculated to a predetermined degree of accuracy and an enthalpy balance is made at the assumed combustion temperature by comparing the enthalpy of the propellant. Iteratively, this method is used to satisfy the combustion temperature, which indicates an enthalpy balance is satisfied. After achieving the correct enthalpy balance, the entropy of the combustion products is calculated at the determined chamber temperature. Then at the assumed exhaust temperature, the composition of the exhaust products are calculated to a predetermined degree of accuracy and an entropy balance is made at the assumed exhaust temperature by comparing the entropy of the combustion products in the chamber and at the nozzle exit. Again, by an iterative method, the exhaust temperature is satisfied such that an entropy balance is satisfied. The enthalpy of the exhaust products is then calculated at the correct exhaust temperature. With this method, the required thermodynamic properties are determined, which allow quantification of the theoretical missile propulsion performance. A summary of calculations is provided in the appendix for ADN/HTPB/Al and AP/HTPB/Al propellant compositions. The theoretical specific impulse obtained for these propellants is determined to be 266.3 and 256.7 seconds, respectively. The propellant properties are summarized below in Table 5.

Table 5. Calculated theoretical propellant properties.

Composition in weight fraction:	ADN 0.6829, HTPB 0.148, Al 0.1691		AP 0.68, HTPB 0.12, Al 0.2	
Density (lb/in ³)	0.0598		0.0649	
Velocity exit (ft/s)	8,570		8,262	
I _{sp} (sec)				
Area ratio (A _e /A _t)	9.4		10.3	
Thrust Coeff. (C _F)	1.612		1.642	
Burn Rate (in/s)				
	Chamber	Exit	Chamber	Exit
Temperature (R)	5,706	3,060	6,120	3,780
Pressure (psia)	1,000	14.7	1,000	14.7
MW (g/mole)	22.423	22.57	25.99	27.67
γ	1.194	1.22	1.192	1.25

From the given derived requirement, it was determined that the solid rocket motor must require both a boost and sustain phase. Moreover, an ideal thrust profile was generated in an attempt to satisfy the performance requirements. Per the above calculated theoretical propulsion performance, APCP is suitable for the boost phase due to its low burn rate value and constraint in missile diameter, and ADN is suitable for the sustainer due to its high burn rate and specific impulse. Figure 12 below shows the schematic of the ideal baseline thrust profile.

**Figure 12. Ideal thrust profile.**

It is also desired to provide a proper grain design to meet the mission requirements. While the motor is in operation, the solid grain burning surfaces recede as more propellant is turned into gas. This burn-back process has a large influence on the resulting thrust-time and motor performance. Depending on the starting grain geometry, a variety of thrust profiles can be achieved. Much of the motor design activity depends on the definition of the grain configuration [32]. Using the above ideal thrust profile, the grain geometry needs to be designed for boost phase to get maximum performance with the same length and diameter of the missile.

To design the grain for the missile, the steps followed were selection and design of grain configuration, evaluation of internal ballistic parameters and analysis of design to meet the ideal thrust profile [30]. For the design, a star shape grain geometry is considered for the missile boost phase to obtain high burn surface area with low sliver

content. The length-to-diameter ratio of the missile and the volumetric fraction makes the internal star shape grain geometry design more promising. Figure 13 shows a schematic of the star pattern grain geometry considered for this design.

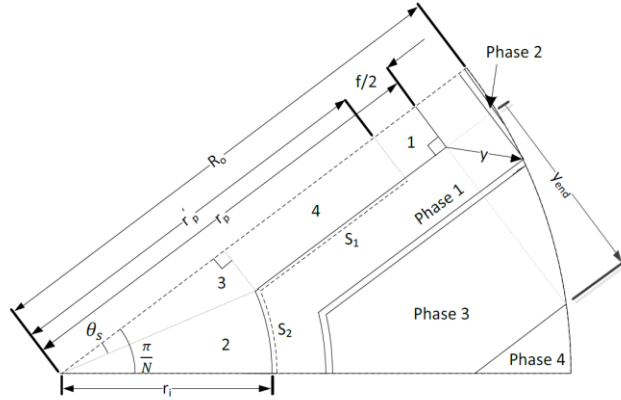


Figure 13. Schematics of star grain geometry showing 1/5th section and nomenclature used for calculations [31].

With this geometry, the burn surface area is calculated during each burn time step [30,31] using the following equations,

$$A_b = (L + y)(S_1 + S_2 + f) \quad (6)$$

$$S_1 = R'_2 - \sqrt{(R_1 + y)^2 - (0.5f + y)^2} \quad (7)$$

$$R'_2 = R_2 - 0.5f \quad (8)$$

$$S_2 = (R_1 + y) \left(\frac{\pi}{N} - \theta \right) \quad (9)$$

$$\theta = \sin^{-1} \left(\frac{0.5f + y}{R_1 + y} \right) \quad (10)$$

$$y_{check} = 0.5 \left(\sqrt{2R^2 - R_2'^2 - R_2'^2} \right) - 0.5f \quad (11)$$

Further, the port area is calculated for only the grain geometry near the nozzle using the following equations,

$$A_{p1} = 2N \left[(0.5f + y)^2 + \frac{1}{2} (R_1 + y) S_2 + \frac{1}{2} (R'_2 - S_1) (0.5f + y) + (0.5f + y) S_1 \right] \quad (12)$$

$$A_{p2} = A_{p1} - 2N \left(\frac{1}{2} R^2 \left[\frac{(0.5f - e + y)1}{R^2} \sqrt{4R^2 \sin^2 \left(\frac{1}{2} \theta' \right) - (0.5f - e + y)^2 - \theta' + \sin(\theta')} \right] \right) \quad (13)$$

$$A_{p3} = A_{p1} - 2N \left(\frac{1}{2} \left[(0.5f + y) \sqrt{4R^2 \sin^2 \left(\frac{1}{2} \theta'' \right) - (0.5f + y)^2 - R^2 \theta'' + R(0.5f + y)} \right] - (0.5f + y)(0.5f + y + R'_2 - R) \right) \quad (14)$$

$$\theta' = \sin^{-1} \frac{0.5f + y}{R} - \sin^{-1} \frac{e}{R} \quad (15)$$

$$\theta'' = \sin^{-1} \frac{0.5f + y}{R} \quad (16)$$

$$e = \sqrt{\frac{1}{(1+p^2)} [R^2(1+p^2) - q^2]} \quad (17)$$

$$p = \cot^{-1} \left(\frac{\pi}{N} \right) \quad (18)$$

For calculation of internal ballistic parameters and analysis of the grain geometry and over-all propulsion performance a MATLAB code is developed. The nozzle mass flow is expressed in terms of the stagnation pressure p_c in the chamber, which must be related to the static pressure in various locations [26]. For this analysis the following equations are utilized,

$$p_1 \simeq p_c \left[1 + \frac{\Gamma^2}{2} F^2 \right] \quad (19)$$

$$p_1 = \left(\frac{\rho_p a K_{eff}}{C_D} \right)^{\frac{1}{1-n}} \quad (20)$$

$$K_{eff} = K \left(1 + \frac{\Gamma^2}{2} F^2 \right) \quad (21)$$

Similarly, the influence of the port area must also be considered in studying the initial pressure rise and the pseudo-equilibrium operation. The initial pressure can be evaluated by,

$$dt = \frac{V_c}{\Gamma^2 c^* A_t p_t (\rho_p c^* K a p_c^{n-1} - 1)} dp_c \quad (22)$$

$$t = \frac{1}{1-n} \frac{V_{\varepsilon(t)}}{\Gamma^2 c^* A_t} \ln \left[\frac{\rho_p c^* K a - p_{c(1g)}^{1-n}}{\rho_p c^* K a - p_c^{1-n}} \right] \quad (23)$$

Finally, the pseudo-equilibrium operations and end burning pressure is calculated using the following relations,

$$\frac{dp_c}{p_c} = \frac{1}{1-n-q} \frac{dk}{K} \quad (24)$$

$$\frac{a C_D V_c}{\Gamma^2 A_t} \frac{dp_c}{dy} = \frac{\rho_p K a}{C_D} - p_c^{1-n} \quad (25)$$

$$\frac{dp_0}{dy} = \frac{p_0}{1-n} \frac{1}{K} \frac{dK}{dy} \quad (26)$$

$$p_c = \left(\frac{\rho_p K a}{C_D} \right)^{\frac{1}{1-n}} \quad (27)$$

D. Structures and Weight Estimates

The structure of the missile must be able to sustain the maximum stresses due to g loads, should withstand the high surface temperatures due to aerodynamic heating and must minimize the weight for the required strength. For a first order approximation of weight from the body geometry dimensions, the chart and the relation from [21] were used.

$$W_L = 0.04 l d^2 \quad (28)$$

where W_L is the launch weight of the missile, l is the length of the missile and d is the diameter of the missile in inches. From the baseline, the initial length of the missile was considered as $l = 236$ in and $d = 13.385$ in. From [21], the structural weight of the missile is around 22% of the total weight, based on the correlations from data on number of missiles.

$$W_s = 0.22W_L \quad (29)$$

The weight of the structure, W_s , should be checked within the baseline weight estimate of 372 lb. The next step is to estimate the thickness of the material required for the structure of the missile. To perform the selection of the material, the load that the material needs to withstand is calculated and then the material thickness is estimated taking a factor of safety for the specific component.

The method to estimate the thickness is to find the minimum thickness required to withstand various loads individually and then to estimate thickness required to handle the effect of combined loading by Root Sum of the Square (RSS) method. The required thickness for various loading for three different materials is tabulated in Table 6. Data from this table are used as a first order estimate of the thickness for the missile and varied as the iterations were proceeded.

Table 6. Thickness of the missile body structure for different materials.

Material	Thickness Required for Individual Loading (in)					Thickness (in)
	Manufacturing Load t_m	Localized Buckling from Bending t_b	Localized Buckling axial compression t_a	Thrust force load t_t	Bending Moment Load t_{bm}	$\sqrt{\left(\frac{(t_m)^2 + (t_b)^2 + (t_a)^2 + (t_t)^2 + (t_{bm})^2}{5} \right)}$
RSA-8009						
AE						
Aluminum	1.09E-01	1.09E-01	2.33E-03	5.09E-03	1.10E-01	2.84E-01
SuperAlloy						
Stainless Steel 310	7.00E-02	2.15E-02	7.76E-04	1.02E-03	2.21E-02	1.15E-01
A-N720						
Oxide Ceramic Matrix Composite	1.15E-01	6.43E-02	2.67E-03	3.03E-03	6.56E-02	2.21E-01

A material selection procedure is now discussed to highlight the same about certain missile components. It is a well-known fact that the nose may burn due to aerodynamic heating and sufficient bluntness is required to sustain the material. A 5% bluntness is added and the iteration is carried out for aerothermal effects and a new aerodynamic database was generated for the updated nose geometry.

The nose casing protects the IR seeker and the electronic sub system throughout the flight trajectory from structural and aerothermal loading. The nose casing can be manufactured using ceramic matrix composite, alumina, such as A-N720 oxide ceramic matrix composite. The high temperature resistant composite maintains strength at temperatures as high as 2000 °F.

The secondary nose, as shown in Fig. 14, protects the avionics and the battery momentarily in the terminal phase after the nose casing is jettisoned. To reduce the complexity of having different material and manufacturing process, 0.12-in thickness of stainless steel 310 is used which could sustain the terminal flight phase [34].

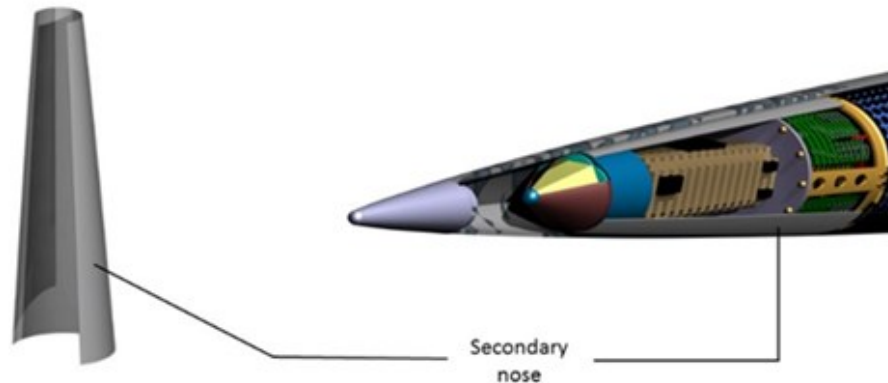


Figure 14. Secondary nose and placement in the missile.

The motor casing must withstand the high pressure generated by the propellant burn. The upper limit of chamber pressure from the operation of the motor is 2,200 psi during the boost phase, hence the casing must be designed in such a way that it can withstand the hoop stress generated from these high chamber pressure. Therefore, the motor casing is manufactured from stainless steel 310 of thickness 0.16 in.

The tail fin geometry was derived from the requirements of the aerodynamics and control. The temperature achieved by the leading edge of the control surface is comparable to the stagnation point temperature at the nose, but with slight offset by the leading edge sweep. The material on the leading edge is capped by high temperature ablative S5000 with the trailing edge made from stainless steel 310.

E. Missile Trajectory

The 3-DOF model for the missile was built in the SIMULINK module of MATLAB 2013b and uses a variable time step, ode45 (Dormand-Prince) solver with a maximum step size of 0.1 seconds and a relative tolerance of 10^{-6} seconds. The model constitutes three major blocks described below to represent the missile engagement scenario.

This block is modeled to represent the motion of the incoming target HGV in space and time. The direction, altitude and downrange of the HGV at the point of launch can be input in the block. Provision is made to represent engagement scenarios with both, non-maneuvering and maneuvering HGVs with a maneuver possible at any stage of the engagement.

This block generates the normal acceleration command on the missile relative to the missile body axis system. A pure proportional navigation system is employed in the model where the magnitude and direction of the demand z-acceleration, a_m , is calculated according to the guidance law,

where, N' is the proportional navigation constant equal to 4 in this case, V_{cl} is the missile closing velocity, and $\dot{\lambda}$ is the rate of change of the line-of-sight (LOS) angle. Schematically, the proportional navigation guidance system can be represented by Fig. 15 presented below,

Figure 15. Proportional navigation schematic [35].

where, R is the range between missile and target, V_m is the interceptor missile velocity, V_t is the velocity of the target, γ_m is the missile flight path (heading) angle, and γ_t is the target flight path angle.

3. Missile and Autopilot

This block is modeled to represent the dynamics of the missile through the time of flight to the target. Integration of input thrust profile and aerodynamic data is made here and the implementation of an atmospheric model to represent derived and calculated properties through the flight path is included. The block consists of 3 major sub-blocks to capture vital aspects of the missile flight dynamics, as elucidated below.

a. Aerodynamics and Equations of motion

The 3 DOF equations of motion are described here in the missile body axis system.

$$\dot{u} = \frac{F_x}{m} - \frac{\dot{m}U}{m} - qw - g \sin \theta \quad (31)$$

$$\dot{w} = \frac{F_z}{m} - \frac{\dot{m}w}{m} + qU - g \cos \theta \quad (32)$$

$$\dot{q} = \frac{M - \dot{I}_{yy}q}{I_{yy}} \quad (33)$$

where, q is the pitch rate, $\dot{u}$ and $\dot{w}$ are missile accelerations in the longitudinal and lateral directions respectively. A reference of the body axis system representation on the missile is presented in Fig. 16. All applied forces act on the center of gravity of the missile. The angle of attack, α , and resultant missile velocity, V , are calculated using

$$\alpha = \tan^{-1} \left(\frac{w}{U} \right) \quad (34)$$

$$V = \sqrt{U^2 + w^2} \quad (35)$$

The main output variables from this block include the instantaneous missile position, missile attitude, missile acceleration, Mach number, pitch and pitch rate, instantaneous missile mass and the static margin.

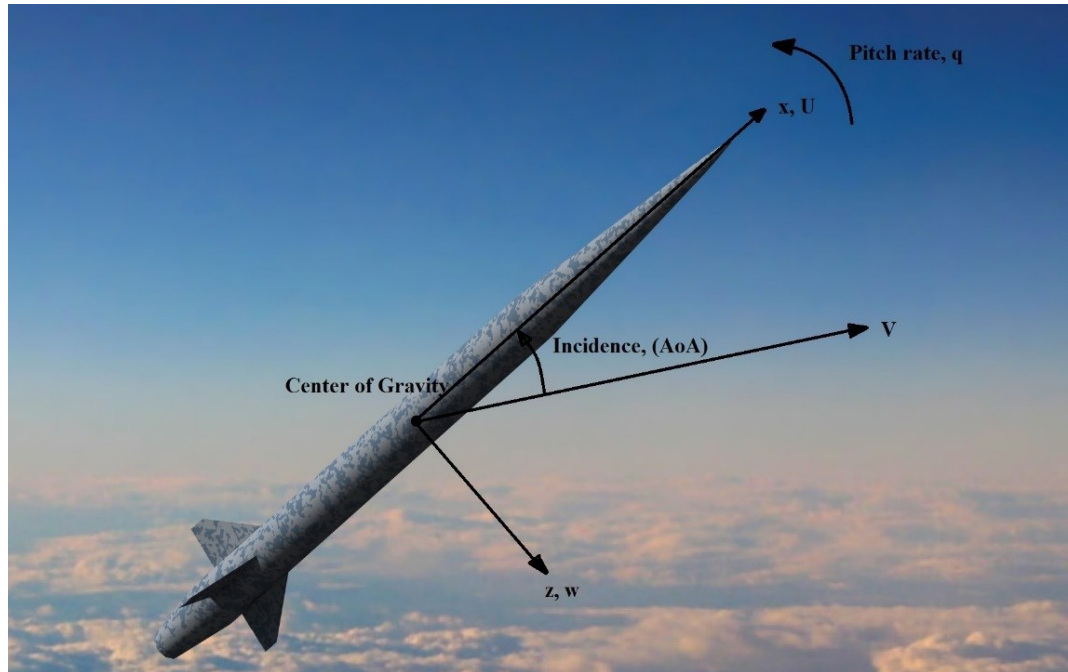


Figure 16. Missile body axis system representation.

In order to determine the necessary axial forces and moments to be input in the equations of motion, they must first be calculated from the force and moment coefficients, respectively, that are derived as functions of a select set of quantities updated at each time interval of the missile flight path. From the Missile DATCOM output quantities, it has been determined that those required set of updated quantities will include the angle of attack (AoA), Mach number, tail fin deflection (δ) and Reynolds number (Re) per foot.

The axial force coefficients (C_a and C_n) and moment coefficients (C_m and C_{mq}) are output for each missile configuration for a range of values of the previously mentioned set of quantities. The swept output is therefore represented as a 4-D structure variable and an interpolation block is implemented to read values of the force and moment coefficients at each flight time interval. Values of the coefficients are linearly interpolated between the closest index values of each independent variable if they do not fall at the precise breakpoint (reference) value in the interpolation scheme.

It is now important to convert the derived force and moment coefficients to the missile body axis system using the simple transformations $C_{a,DATCOM} = -C_{x,missile}$ and $C_{l,DATCOM} = -C_{z,missile}$. Similarly, the lift and drag coefficients are calculated as described below.

$$-C_l = C_{z,missile} \cos \alpha - C_{x,missile} \sin \alpha \quad (36)$$

$$-C_d = C_{z,missile} \sin \alpha + C_{x,missile} \cos \alpha \quad (37)$$

Furthermore, the lift, drag and axial forces are calculated as,

$$L_f = \frac{1}{2} \rho V^2 S_{ref} C_l \quad (38)$$

$$D_f = \frac{1}{2} \rho V^2 S_{ref} C_d \quad (39)$$

$$F_x = \frac{1}{2} \rho V^2 S_{ref} C_x \quad (40)$$

$$F_z = \frac{1}{2} \rho V^2 S_{ref} C_z \quad (41)$$

Note that required terms in the equations of motion are the normal and axial forces and not the lift and drag. The moment about the CG is now calculated as,

$$M = \frac{1}{2} \rho V^2 S_{ref} L_{ref} C_m \quad (42)$$

where, $S_{ref} = \pi D_{missile}^2 / 4$ and L_{ref} are the reference area and length from Missile DATCOM. The moment from the dynamic stability derivative is calculated as,

$$C_{m,cmq} = \frac{C_{mq,DATCOM} q L_{ref}}{2V} \quad (43)$$

where it must be noted that Missile DATCOM outputs $C_{mq} \text{ deg}^{-1}$ and must be converted to rad^{-1} to stay consistent with the units. The total contribution to the moment coefficient would now be,

$$C_m = C_{m,DATCOM} + C_{m,cmq} \quad (44)$$

The moment of inertia term is calculated based only on the cylindrical missile body since the nose contributed minimally to the total missile moment of inertia [21]. Therefore,

$$I_{yy} = \left(\frac{W d^2}{g} \right) \left(\frac{1}{16} + \frac{l}{12 d^2} \right) \quad (45)$$

where W/g is the instantaneous missile mass at every flight time interval. The missile static margin (SM), in calibers, is calculated as,

$$SM = \frac{X_{CP} - X_{CG}}{L_{ref}} \quad (46)$$

where, X_{CP} is the center of pressure and is swept across the instantaneous AoA, Mach number, δ and Re . The atmospheric block implemented here is the 1976 Committee on Extension to the Standard Atmosphere (COESA) model, where the temperature, density, pressure and speed of sound (all output in metric units) are a function of the missile altitude. The absolute viscosity is then calculated from Sutherland's law as a function of temperature.

$$\mu = 2.27 \times 10^{-8} \left(\frac{T^{1.5}}{T + 199} \right) \quad (47)$$

The temperature is in °R and the unit of μ is lbf · s/ft². The Reynolds number (Re) is then calculated per unit foot.

b. Missile Sensor (Accelerometer/Rate Gyro)

This block senses the normal and axial missile acceleration and measures the missile pitch rate. It is assumed that the accelerometer is located 0.33 ft fore of the average missile CG and hence, the normal missile acceleration is corrected for by using the equation below.

$$A_{z_m} = A_{z_{CG}} - 0.33\dot{q} \quad (48)$$

c. Missile Autopilot

The autopilot inputs missile flight parameters from the Aerodynamics and Missile Sensor blocks to output the required fin deflection, δ , in order to maintain an intercept course with the incoming HGV.

An acceleration control system is employed in the autopilot design of the missile as depicted in Figure 17. The outer negative feedback acceleration loop is controlled via a proportional gain K at all flight conditions while the inner positive feedback pitch rate loop is proportional-integral (PI) controlled via gains K and K_i .

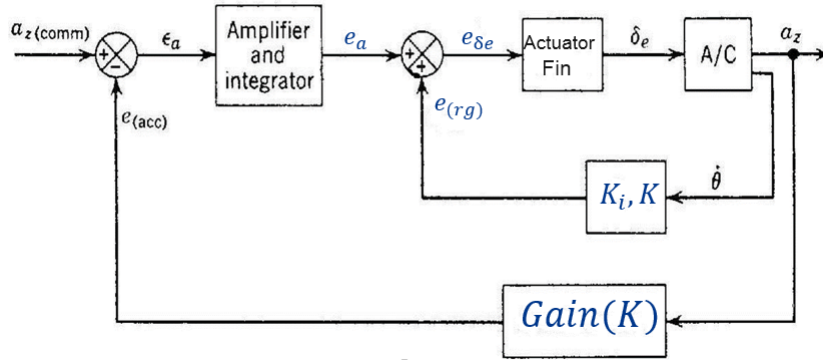


Figure 17. Acceleration control system [36].

The proportional gain on the outer loop and the integral gain on the inner loop are scheduled based on the instantaneous value of the missile static margin at every point in the flight. Since the outer loop input, δ_e , varies directly with the output a_z the sign on the summer is negative to calculate an error, ϵ_a . A converse argument follows to determine the summer sign in the case of the inner loop where the input δ_e varies inversely with the pitch rate $\dot{\theta}$.

F. Performance Assessment

As indicated in the design methodology flowchart of Fig. 4, once the aerodynamics, propulsion, structures and weights, and trajectory data are determined, the missile system must be evaluated to determine if the configuration satisfies the required performance. For this particular design step, a performance checklist is utilized to address the major design disciplines to ensure performance aspects within each discipline are satisfied. The crucial performance requirements are those derived based on the problem description. These include the time-to-target and achievable

missile range. Similarly, regarding the aerodynamics, it is required that the supplied thrust from the propulsion system be larger than the aerodynamic drag over the full course of the missile trajectory. Additionally, it is desired to have the missile operate statically stable with an acceptable degree of maneuverability. For the propulsion, it is required that the burn time be equal to or longer than the time-to-target to avoid the missile coasting to the incoming target. Further, regarding the missile structure, it is required that the missile structure sizing be able to handle the aerodynamic loadings imparted on the vehicle during flight. This includes the axial compression, bending stress, and buckling stresses imparted on the structure during the flight trajectory. Additionally, due to the missile flight Mach number, aerodynamic heating must be considered at the stagnation points on the missile. The primary missile regions affected by severe frictional heating are the nose tip and control surface fin leading edges. Regarding the missile trajectory, it is desired to avoid control surface deflections during flight that exceed a desired threshold. This is a constraint on the available power supply and the required power demand for large control surface deflection at high Mach numbers. Further, with large control surface deflection demands, the missile simultaneously experiences large increases in aerodynamic drag and subsequently decreases the obtainable range. Finally, it must be verified that the missile can intercept an approaching hostile HGV outside of the 7 km protection radius. Table 7 below summarizes the missile performance checklist used to properly assess the missile capability for each design iteration.

Table 7. Missile performance checklist to satisfy counter HGV requirements.

Performance Requirement	Metric
Time-to-target	~ 60 sec
Missile Range	~ 40 -50 miles
Aerodynamic Drag	Thrust > Drag
Static Margin	$0 < SM < 1.0$ cal.
Burn Time	$t_{burn} \geq t_{intercept}$
Structure	$\sigma_{allow} > \sigma_{comp}, \sigma_{bend}, \sigma_{buckle}$
Aerodynamic Heating	$T_{allow} > T_{nose}, T_{fin}$ at stagnation point
Fin Deflection	$\delta_{allow} > \delta_{required}$
Intercept Range	$R_{Intercept} > 7$ km (4.35 miles)

After applying the above design methodology for the missile baseline aerodynamics, propulsion, structure and weight, several trajectories were analyzed to assess the baseline missile performance against the checklist detailed in Table 10. Since, a weight correlation was utilized to estimate the weight of the baseline missile configuration, the structural design requirements will not be checked for the baseline as a material sizing was not conducted. However, each subsequent iteration will entail a material selection and sizing to handle the aerodynamic loading and heating computed from the current missile trajectory. Applying this checklist to the baseline missile performance yields the results shown in Table 8 below.

Table 8. Summary of baseline missile performance.

Performance Requirement	Metric	Obtained	Pass/Fail
Time-to-target	~ 60 sec	74.2 sec	Fail
Missile Range	~ 40 -50 miles	24 miles	Fail
Terminal Mach Number	~ 6	3.5	Fail
Aerodynamic Drag	Thrust > Drag	✓	Pass
Static Margin	$0 < SM < 1.0$ cal.	~ 2	Fail
Burn Time	$t_{burn} < t_{intercept}$	✗	Fail
Structure	$\sigma_{allow} > \sigma_{comp}, \sigma_{bend}, \sigma_{buckle}$	---	---
Aerodynamic Heating	$T_{allow} > T_{nose}, T_{fin}$ at stagnation point	---	---
Fin Deflection	$\delta_{allow} > \delta_{required}$	✓	Pass
Intercept Range	$R_{Intercept} > 7$ km (4.35 miles)	✓	Pass

From these results, it is evident that the baseline missile fails to satisfy the approximate 60 second time-to-target. Additionally, the baseline missile is only able to obtain about half the required range to sufficiently intercept an incoming threat HGV. Likewise, the terminal Mach number achieved is only 3.5 as opposed to the approximate requirement of 6. Further, the static margin is approximately a caliber above that which would allow the missile to maintain adequate maneuverability, which would likely render the missile incapable of defending against a maneuvering HGV. With these results, it is obvious that several design iterations are required to ultimately satisfy the mission requirements. These subsequent iterations will necessitate the resizing of the missile configuration,

propulsion system, and weight in order to properly converge the design towards a feasible and capable missile system.

IV. Missile System Design Iterations

Below is a brief summary of the analysis and decision making process used to successfully apply the iterative design procedure outlined in the design methodology.

At iteration 2, the baseline missile trajectory was used to establish aerodynamic loadings subjected to the missile structure during flight. Maraging steel was selected as structure material and sized accordingly to withstand stresses from axial compression, bending, and buckling. With this data, a new weight estimate was determined based on resultant structure sizing and weight.

Two iterations later, the tail area was reduced from 1.59 ft^2 to 1.23 ft^2 per control surface in order to reduce the wetted tail surface area and effectively shift the static margin towards the center of gravity of the missile. This provides additional maneuverability for the missile during the terminal phase in the event the system encounters an incoming maneuvering HGV. Additionally, this reduces the hinge moment requirement and control surface deflection, which reduces the overall drag throughout the course of the missile trajectory.

At the 8th iteration, the diameter of the missile was reduced from 1.12 ft to 1.05 ft and the missile center body and nose were reduced by a combined 3.28 ft. This caused a slight penalty regarding an increase in aerodynamic drag due to a marginal increase in the missile fineness ratio, however the placement of the new center gravity provided better stability performance regarding the static margin.

At the 10th iteration, the propellant mass was increased to lengthen the overall burn time, and the thrust profile was adjusted to that obtained using the burn area calculations. Figure 18 is a schematic of the designed booster stage, which consisted of 3 distinct grain geometries to lengthen the boost burn time by approximately 3 seconds and simultaneously provide a higher thrust.

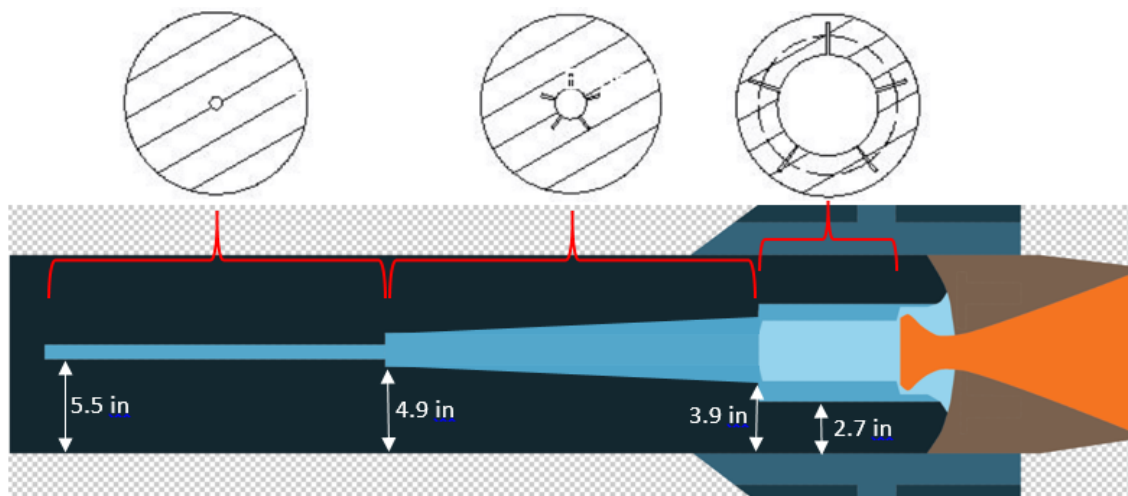


Figure 18. Schematic of boost phase showing three sub phases.

The final rocket motor design, propellant burn profile, thrust profile are shown in Figures 19, 20 and 21 below.



Figure 19. Schematic of complete solid rocket motor used on missile.

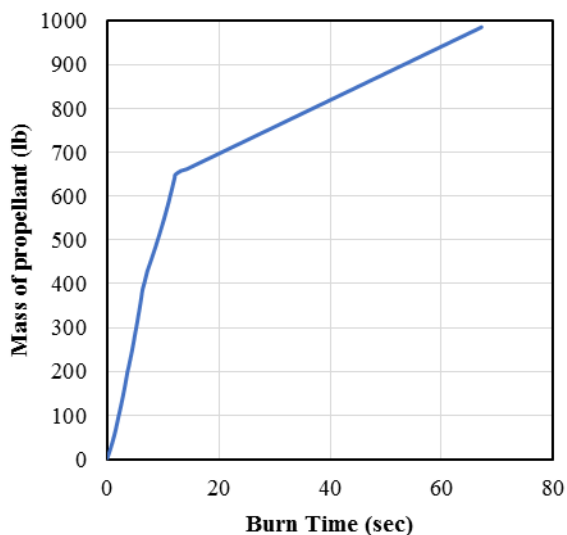


Figure 20. Propellant mass burnt with time.

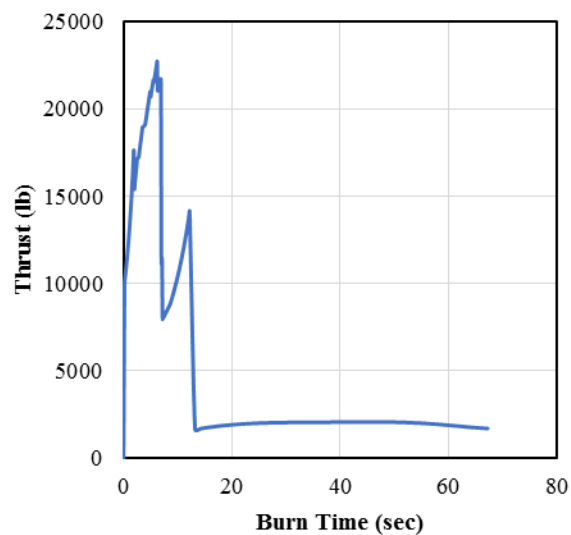


Figure 21. Final thrust profile.

At iteration 11, a detailed heat transfer analysis was conducted to determine the effects of aerodynamic heating at the stagnation points on the nose tip and leading edges of the fins during the flight trajectory. Figure 22 below shows that the stagnation temperature exceeds the maximum service temperature of the maraging steel selected for the nose and fin design.

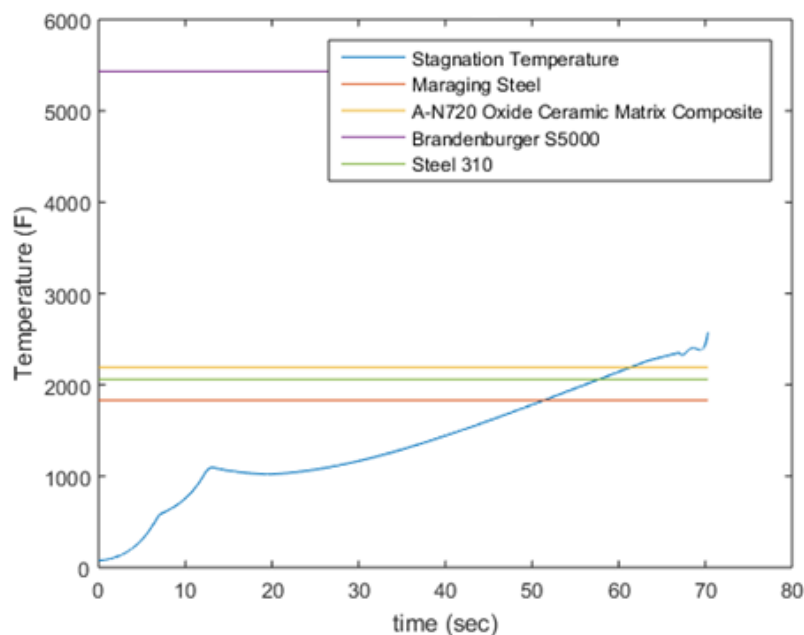


Figure 22. Stagnation point temperature on the nose and fin.

Therefore, as part of this iteration, a detailed nose and fin design was carried out such that the leading edge material could withstand the Mach 6 environment subjected on the missile near and during the terminal phase. For the nose, a 5% bluntness was added and the tip base material was changed to 310 stainless that is then coated with S5000, a high temperature ablative material. Figure 23 below provides a schematic of the nose tip design.

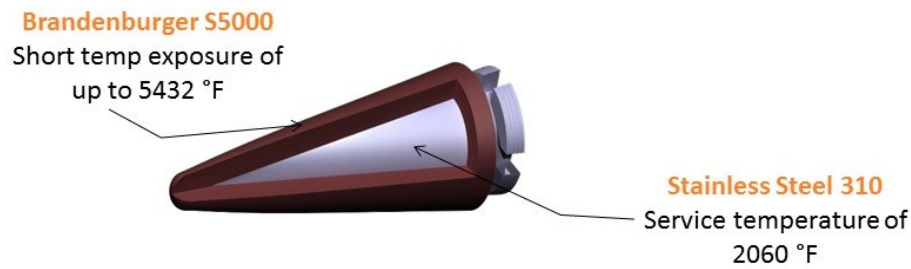


Figure 23. Nose tip with high temperature ablative layer over 310 stainless steel.

Likewise, for the leading edge of the control surface fins, the base material was changed to 310 stainless that is then coated with S5000. Figure 24 below provides a schematic of the fin design.



Figure 24. Fin design with high temperature ablative layer over 310 stainless steel.

Iteration 12 saw the nose shape change from a tangent-ogive to a $\frac{3}{4}$ -power series for increased internal volume to properly house the missile subsystems.

Iteration 13 consisted of a final weight estimate based on the final propulsion system and structure design. Table 9 below is the performance assessment checklist applied to the final design. Notice, that each the derived performance requirements for a single stage missile are satisfied.

Table 9. Final design performance checklist summary.

Performance Requirement	Metric	Obtained	Pass/Fail
Time-to-target	~ 60 sec	65.6 sec	Pass
Missile Range	~ 40 -50 miles	46.5 miles	Pass
Terminal Mach Number	~ 6	5.94	Pass
Aerodynamic Drag	Thrust > Drag	✓	Pass
Static Margin	$0 < SM < 1.0$ cal.	~ 0.4	Pass
Burn Time	$t_{burn} \geq t_{intercept}$	✓	Pass
Structure	$\sigma_{allow} > \sigma_{comp}, \sigma_{bend}, \sigma_{buckle}$	✓	Pass
Aerodynamic Heating	$T_{allow} > T_{nose}, T_{fin}$ at stagnation point	✓	Pass
Fin Deflection	$\delta_{allow} > \delta_{required}$	✓	Pass
Intercept Range	$R_{Intercept} > 7$ km (4.35 miles)	✓	Pass



Figure 25. Design iteration process to approach final missile design.

The final missile resulting from the above detailed iterative process is named ARX, and the missile specifications are elaborated in Table 10 below.

Table 10. ARX missile system specifications.

Component	Dimension
Length Nose	5.99 ft
Diameter of Nose	1.05 ft
Length of Centerbody	10.97 ft
Diameter of Centerbody	1.05 ft
Boat Tail Length	1.00 ft
Boat Tail Angle	5°
Airfoil	Double-wedge
Airfoil Thickness	5 %
Root Chord	1.75 ft
Taper Ratio	0.25
Semi-Span	1.5 ft
Leading Edge Sweep	50°
Fin Area	1.07 ft ²
Reference Area	0.87 ft ²
Launch Weight	1,796 lbs
Propellant Weight	985.7 lbs
Motor Burn Time	67.2 s

V. Detailed Missile System Design and Considerations

A. Missile Performance against HGVs

This section highlights important missile performance metrics over a nominal engagement scenario of the final missile configuration that was chosen. Major performance parameters considered are the missile range, Mach number, thrust-to-drag ratio, missile accelerations and the closeness of the normal acceleration with the commanded acceleration from the guidance system, lift, drag, normal and axial force coefficients, dynamic pressure, Reynolds number per foot, angle of attack and tail fin deflection (Figure 27-36). All performance metrics are shown over the entire flight trajectory showcasing a nominal, non-maneuvering target scenario.

The aerothermal heating is an important aspect in the performance of the structural design of the missile. It is not the steady state temperature that plays a major role in the structural design, but the transient heat transfer that dominates the structure aerodynamic heating during the missile flight. The flight time of missile is low compared to hypersonic aircraft, which reduces the complexities but still an area that must be addressed for any high-speed flight vehicle [37]. For calculating the aerothermal effects, an in-house MATLAB code is developed based on impact methods [38] for calculating the flow condition on the missile, which further utilizes the Eckert theory [39] for calculating the temperature on the surface of the missile.

The trajectory data is loaded for every single increment in Mach number for altitude and time values. Then, based on the heat transfer coefficient and the time exposed, the wall temperature is calculated at each time step. This permits real-time and high fidelity evaluation of the surface temperature distribution with minimal computational expense.

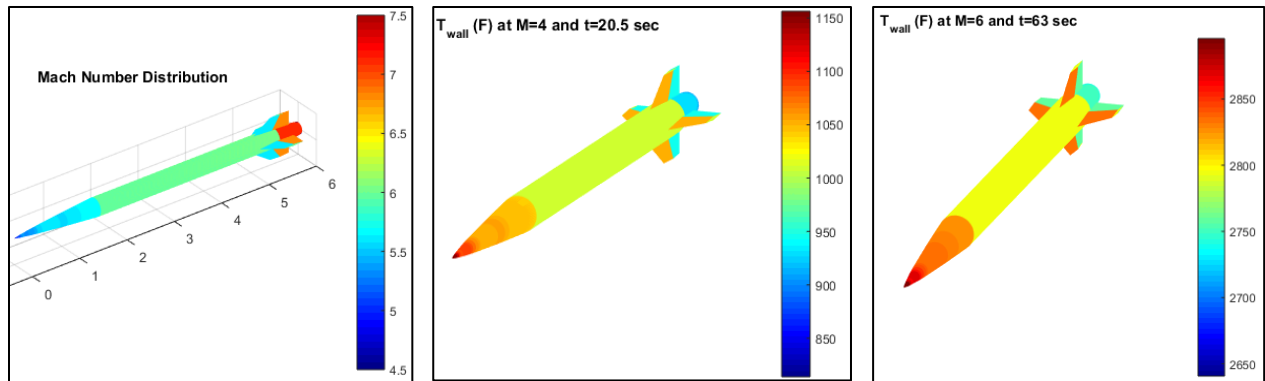


Figure 26. Aerodynamic heating analysis conducted with aerothermal MATLAB code.

Alongside the surface temperature heat transfer, the stagnation point temperature was plotted for the entire trajectory using a recovery factor of 0.89. As shown in Fig. 26, the temperature of the nose is well within the range of stainless steel 310 till 60 secs of flight time, but increases beyond that point. The steady rise in temperature is due to the increase in hypersonic Mach number after 20 secs of flight. Therefore, to keep the structure within the safe limit, regions that exceed the critical temperature were replaced by high temperature resistant ablative material.

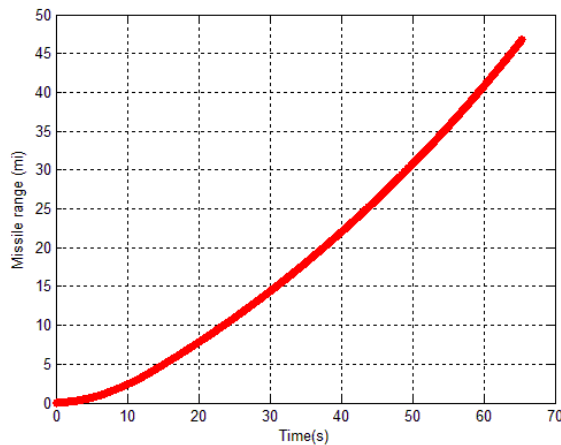


Figure 27. Missile range (mi) vs. time (s).

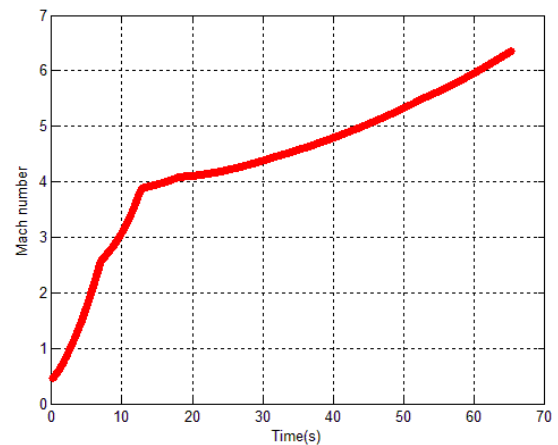


Figure 28. Missile Mach number vs. time (s).

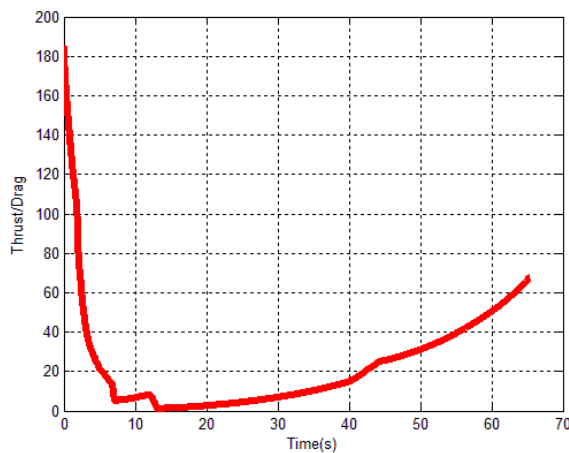


Figure 29. Thrust-to-drag vs. time (s).

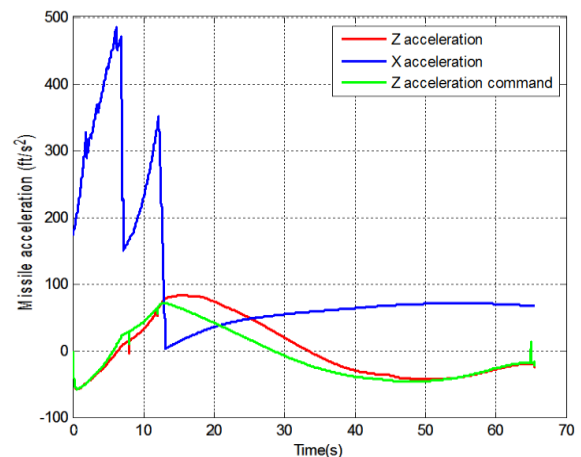


Figure 30. Missile acceleration vs. time (s).

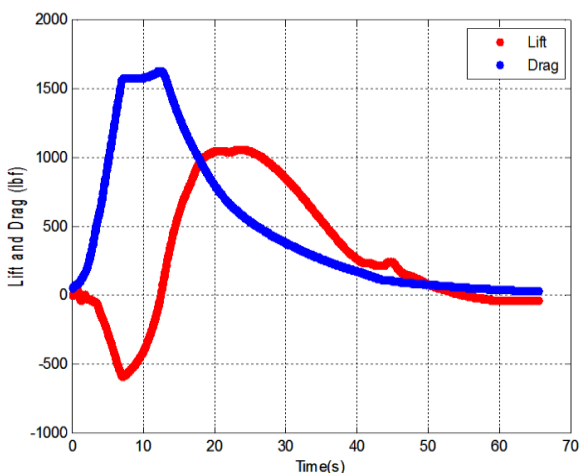


Figure 31. Missile lift and drag vs. time (s).

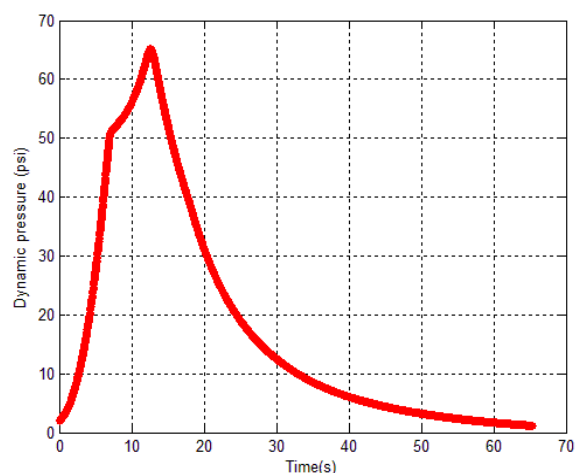


Figure 32. Missile dynamic pressure vs. time (s).

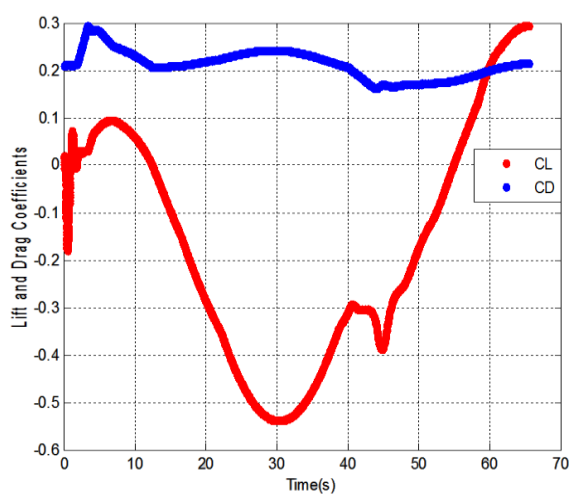


Figure 33. Lift and drag coefficient vs. time (s).

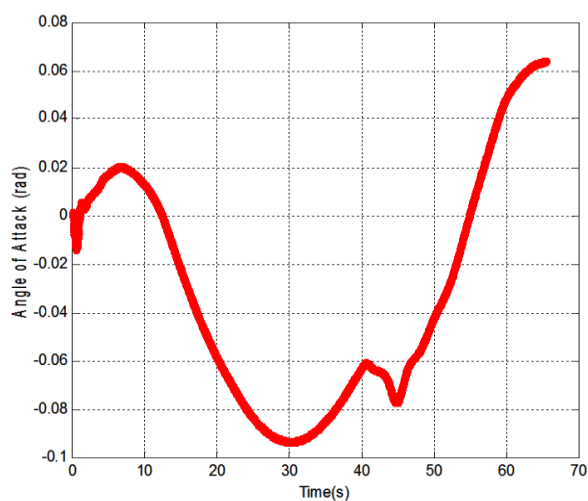


Figure 34. Angle of attack vs. time (s).

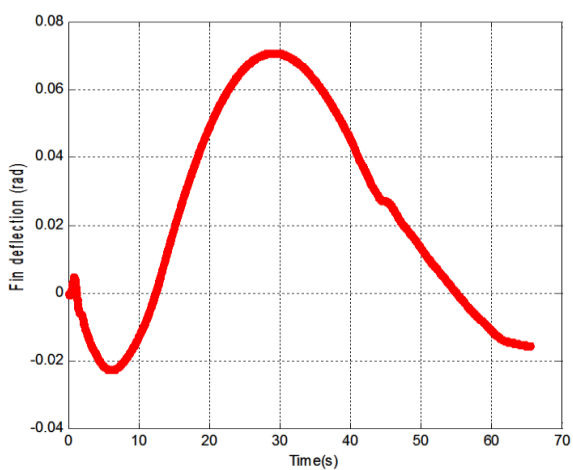


Figure 35. Fin deflection vs. time (s).

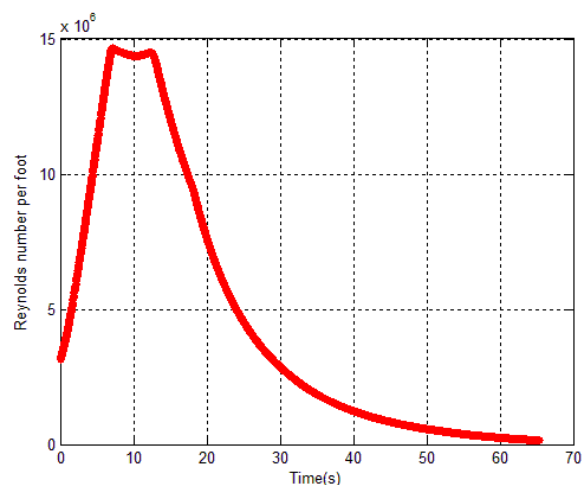


Figure 36. Reynolds number per foot vs. time (s).

Lastly, Fig. 37 demonstrates the ARX's capability to defend against both non-maneuvering and maneuvering target HGVs that are approaching with Mach numbers ranging from 12–15 at altitudes varying from 120–140 kft. This demonstrates that the final missile design is capable of sufficiently achieving the protection goals as demanded in the formulated requirements.

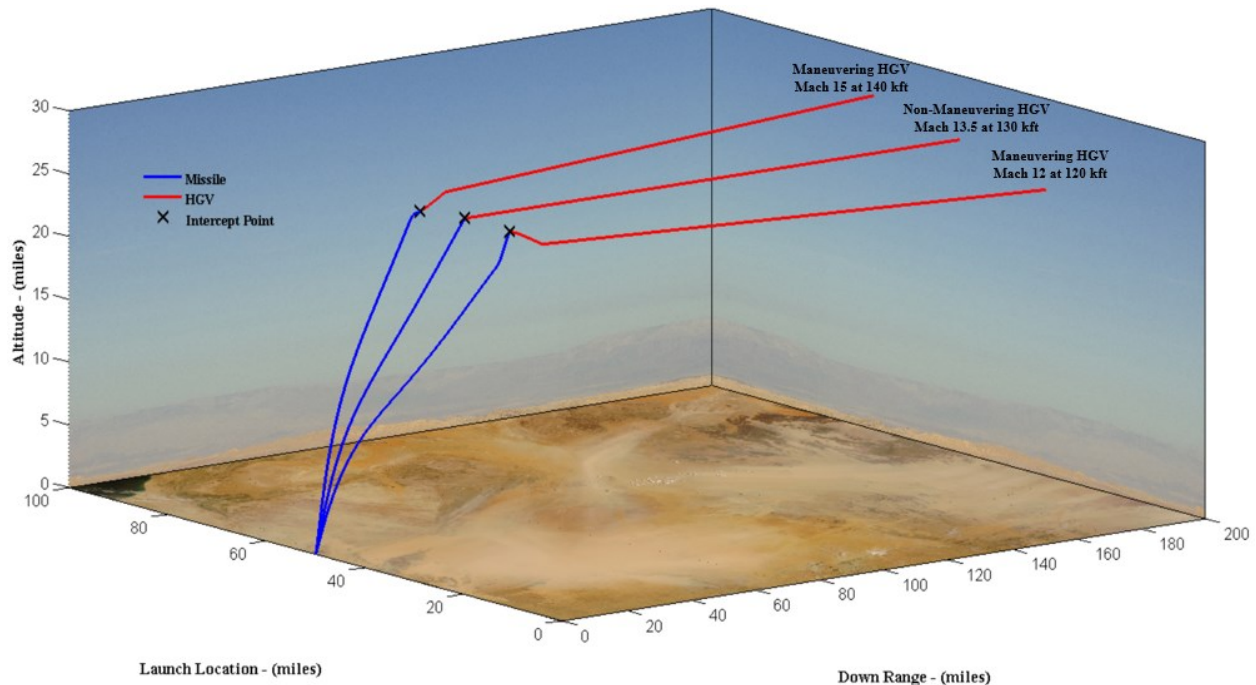


Figure 37. ARX missile defense system intercepting 3 HGVs approaching at various Mach numbers and altitudes.

B. Missile Subsystem Components and Inboard Layout

The avionics segment of the missile subsystem components primarily include the seeker and the inertial measurement unit (IMU). The choice of the missile seeker depends on the stated mission requirements and the seeker operating conditions. A passive imaging staring IR seeker with an electronic (strapdown) tracking and stabilizing mechanism will be featured in the ARX missile. Distinctive advantages of vital seeker features include passive imaging that offers advantages over an active seeker in terms of lower noise and observability. Only the total energy emitted off the target falls under consideration and not the temperature/size/reflected intensity as required in active imaging technology [21]. The seeker is designed to be a rotating, actively cooled model and operates outside the cloud cover envelope in the terminal phase of flight. A staring Focal Plane Array (FPA) whose advantages include lighter weight, longer detection range and less sensitivity to clutter [21]. It can be assumed that sufficient maturity and development of staring FPA technology is reasonable at the time of system deployment and operation. An electronic tracking mechanism is deployed to lower cost, reduce weight and radome error in comparison with a gimbaled seeker mechanism. Considering the mission profile likely to be executed in terms of the relative positions and orientations of the target and missile at the point of intercept (head-on versus tail-chase scenarios), the need for a large field of regard (FOR) is negated by the limited off-boresight tracking requirements.

The IMU houses the sensors required to determine the position, orientation and acceleration of the missile. The IMU configuration includes a magnetometer, rate gyro and accelerometer on each of the three missile body axes. The magnetometer measures tracked movements and combines it with the timing from the synchronized master timing clock, the rate gyros determine the attitude and angular rates about the CG of the missile and the accelerometers measure acceleration along missile axes based on their location relative to the missile CG.

The ARX missile includes a blast fragment subsystem as a means to enhance kill probability. The fragment warhead properties must be estimated to effectively model the blast fragmentation. The key aspects of a fragment warhead are the fragment density and velocity decay. To arrive at these values, various properties of the fragment

must be estimated. Several explosives were considered namely, RDX B, RDX C-3, Tritonal, TNT, Torpex and Pentolite. Limited by the availability of data and the key parameter of explosive known as gunnery constant, RDX C-3 is chosen as the explosive charge [40].

A detailed inboard layout of the ARX missile is shown in Figure 38 below. The schematic labels the major missile subsystem components mentioned above, and serves to highlight their locations in the missile.

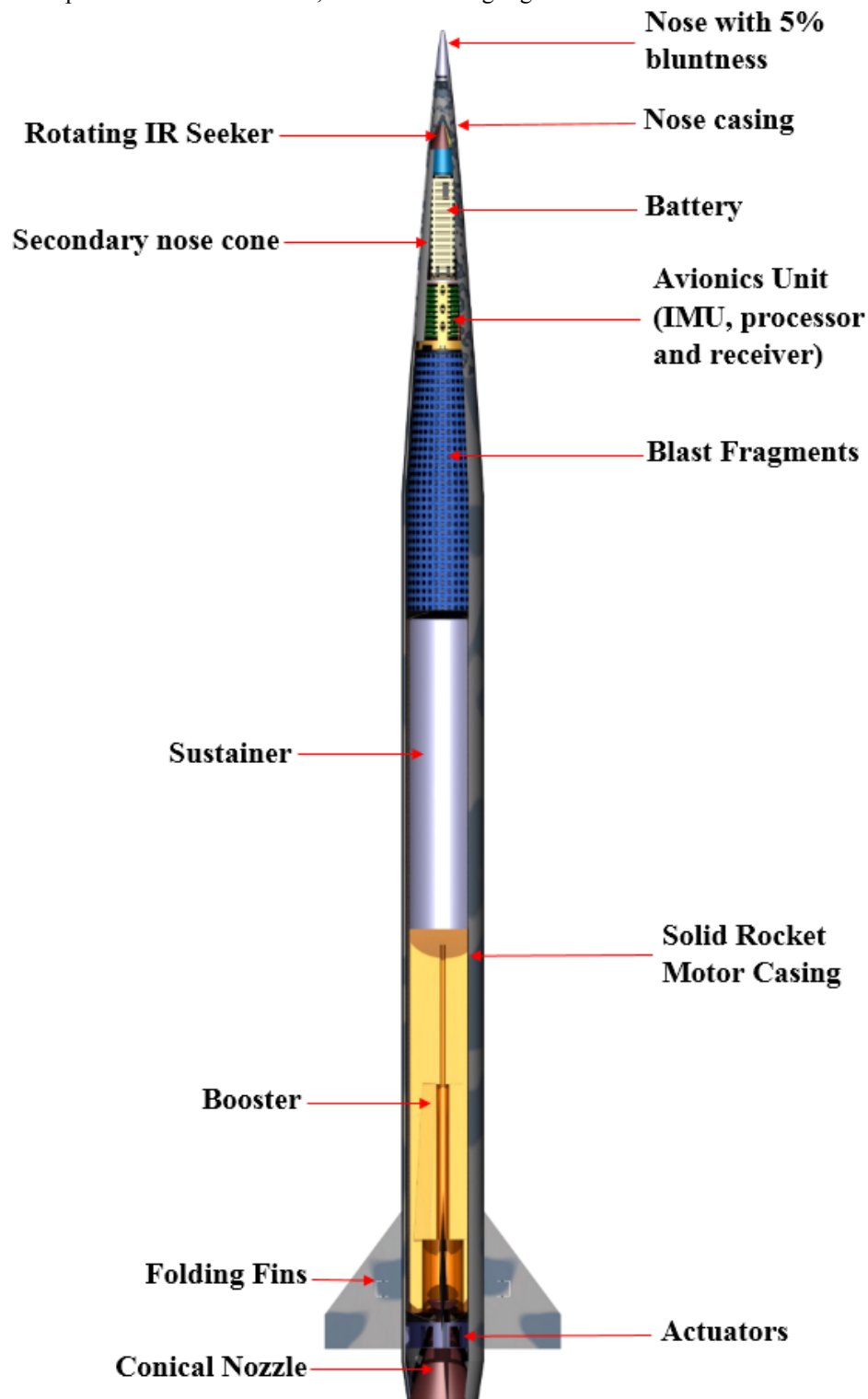


Figure 38. Inboard profile view of the ARX missile system.

VI. Conclusion

A detailed conceptual analysis and design of a single-stage, hit-to-kill missile system has been developed to counter a HGV threat during cruise as stated in the mission requirements. Elaborate analysis tools have been developed to support an iterative design process to accomplish all derived requirements. Major performance metrics for a nominal case that satisfied the requirement goals set out include a time-to-target of approximately 65 seconds, an intercept Mach number of approximately 6, a missile range between 40 to 50 miles and a static margin between 0 and 1 calibers. Practical engagement scenarios involving maneuvering and non-maneuvering targets operating over the required range of altitudes and velocities were simulated to demonstrate the missile system's capability to intercept the HGV(s) successfully.

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